Adjoint Sampling with Stein-Based Variance Reduction — Full Results

> **Generated:** 2026-04-04 (DW4 + LJ13 complete) **Repo:** `adjoint_samplers`
> **Methods:** 12 post-hoc enhancement methods evaluated across 2 molecular systems

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Table of Contents

1. Overview
2. Systems Under Test
3. Methods
4. DW4 — Double Well 4 Particles (8D)
5. LJ13 — Lennard-Jones 13 Particles (39D)
6. Cross-System Comparison
7. Ablations
8. Key Takeaways

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1. Overview

We train Adjoint Sampling with Bridge Splitting (ASBS) to sample from Boltzmann distributions of molecular systems, then apply **12 post-hoc enhancement methods** to reduce estimator bias and variance.

Two observable targets: - **Energy:** $\langle E \rangle_p$ (primary, all methods) - **Interatomic distance:** $\langle \bar{d} \rangle_p$ (methods 10–12 only)

Evaluation protocols: - **Full-sample protocol:** 10 seeds × 4 sample sizes ($N \in \{100, 500, 1000, 2000\}$), sample mean evaluation - **Bootstrap protocol:** 3 seeds × 200 bootstrap trials (subsample size 200), per-observable - Ground truth: sample mean of reference samples from the target distribution $\pi \propto e^{-E(x)}$ - All metrics reported as mean ± std across seeds

Bias-variance coupling theorem:

$$|\text{Bias}| \leq \sqrt{C \cdot \text{Var}_{q_\theta}[f + \mathcal{A}_p g]}$$

Optimizing the control variate g for variance reduction automatically shrinks the bias. All Stein CV methods exploit this — they differ only in how they parameterize and train g.

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2. Systems Under Test

System	Particles	Spatial Dim	Total Dim	Energy	Reference Samples
DW4	4	2D	**8**	Pairwise double-well	10,000
LJ13	13	3D	**39**	Lennard-Jones + harmonic	1,000

DW4 — $E_{\mathrm{DW}}(d) = 0.9(d-4)^4 - 4(d-4)^2$ pairwise over 4 particles in 2D. Fast iteration, good for debugging all enhancements.

LJ13 — $E_{\mathrm{LJ}}(r) = \varepsilon\left[\left(\frac{r_{\min}}{r}\right)^{12} - 2\left(\frac{r_{\min}}{r}\right)^{6}\right]$ with $\varepsilon = 1, r_{\min} = 1$ and external harmonic confinement. 13 particles in 3D with COM constraint. Medium complexity.

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3. Methods

3.1 Method Catalogue

#	Method	Estimator / Mechanism	Complexity	Bias Fix?	Var Fix?
1	**Vanilla ASBS**	Sample mean $\frac{1}{N}\sum f(x_i)$	$O(N)$	—	—
2	**RKHS Stein CV**	Optimal RKHS control variate via kernel ridge regression	$O(N^3)$	✓ (coupling)	✓
3	**Antithetic Sampling**	Paired trajectories with negated Brownian noise	$O(N)$	—	✓
4	**MCMC Correction**	K Metropolis-Hastings steps on terminal samples	$O(KN)$	✓	—
5	**MCMC + RKHS Stein CV**	MCMC first, then kernel Stein CV	$O(KN+N^3)$	✓✓	✓
6	**Generator Stein CV**	Learned drift $b_\theta(x,1)$ in place of $s_p(x)$	$O(N^3)$	✓ (coupling)	✓
7	**Neural Stein CV** (PDE loss)	MLP g_ϕ trained on $\|\nabla_x h\|^2$	$O(BdT)$	✓ (coupling)	✓

#	Method	Estimator / Mechanism	Complexity	Bias Fix?	Var Fix?
8	**EGNN Stein CV** (PDE loss)	Equivariant g_ϕ on PDE loss $\|\nabla_x h\|^2$	$O(Bn^2LT)$	✓ (coupling)	✓
9	**RBF Collocation CV**	g expanded in Gaussian RBF basis, single least-squares solve	$O(NdM + M^3)$	✓ (differentiated PDE)	✓
10	**Equivariant Stein CV** (Var loss)	EGNN + LayerNorm, $\min \mathrm{Var}[f + T_\nu g]$ with early stopping	$O(Bn^2LT)$	✓ (coupling)	✓
11	**ISM + Stein CV**	Implicit Score Matching diagnostic + variance-loss CV	$O(BdT)$	✓ (coupling)	✓
12	**Score-Informed CV**	$g = \alpha \cdot g_{\text{init}} + g_{\text{res}}$, ISM-derived g_{init}	$O(BdT)$	✓✓ (score-aware)	✓✓

3.2 Two Loss Functions for Stein CV Training

All neural Stein CV methods train g_ϕ so that $h = f + T_\nu g$ has minimal variance. Two losses are used:

Aspect	PDE Loss (methods 7–9)	Variance Loss (methods 10–12)
Objective	$\min \|\nabla_x (f + T_\nu g)\|^2$	$\min \mathrm{Var}[f + T_\nu g]$
Requires ∇f?	Yes	No
Validation	None	Train/val split + early stopping
Grad clip	10.0	5.0
g-network	MLP or EGNN (generic)	EGNN + LayerNorm (equivariant)
Observable-agnostic?	No (needs differentiable f)	Yes

3.3 Method Details

Methods 7–9 (PDE-loss Stein CVs): - **Neural Stein CV** uses a generic MLP ignoring particle structure. - **EGNN Stein CV** exploits E(3) symmetry of molecular systems, constraining g_ϕ to physically meaningful vector fields. - **RBF Collocation CV** avoids neural training entirely — expanding g in a Gaussian RBF basis turns the PDE into a **linear least-squares** problem. One matrix solve, no epochs, no gradient instability.

Methods 10–12 (variance-loss Stein CVs): - **Equivariant Stein CV (Var loss)** uses the same EGNN architecture but optimizes $\mathrm{Var}[h]$ directly instead of $\|\nabla h\|^2$. This doesn't require ∇f, supports early stopping with train/val split, and uses LayerNorm

for stability. - **ISM + Stein CV** first trains a score model $s_\phi \approx \nabla \log q_\theta$ via Implicit Score Matching, providing diagnostics on sampler quality. The CV itself uses variance loss with the equivariant architecture. - **Score-Informed CV** decomposes $g = \alpha \cdot g_{\text{init}} + g_{\text{res}}$ where $g_{\text{init}} = -(\nabla E + s_\phi)$. When $q_\theta \approx \pi$, $g_{\text{init}} \approx 0$ and $T_\nu g_{\text{init}} \approx 0$, so the residual g_{res} only corrects the remaining error.

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4. DW4 — Double Well 4 Particles (8D)

4.1 Training

Parameter	Value
Experiment	`dw4_asbs`
Epochs	5000
NFE	200
$\sigma_{\max}$ / $\sigma_{\min}$	1.0 / 0.001
Source	Harmonic (scale=2)
Checkpoint	`results/local/2026.03.31/152919/checkpoir`

4.1.1 Ground Truth

Computed from 10,000 reference samples drawn from $\pi \propto e^{-E(x)}$ (`data/test_split_DW4.npy`):

Statistic	Value
Mean	**-22.4504**
Std	1.9015
Median	-22.7987
[5th, 95th] %ile	[-24.9067, -18.8260]
[Min, Max]	[-25.7486, -11.8381]

4.2 Full-Sample Results — Energy (10 seeds)

N = 2000:

Method	Mean Energy		Error	
Ground Truth	**-22.4504**	0	—	—
Vanilla ASBS	-22.4093 ± 0.0443	0.0456	1.92e-03	1.000
RKHS Stein CV	-25.0793 ± 0.1873	2.6289	1.45e-03	0.759
Antithetic	-22.3934 ± 0.0387	0.0592	1.32e-03	0.688
MCMC (K=10)	-22.4084 ± 0.0471	0.0467	1.92e-03	—
MCMC + RKHS CV	-24.8512 ± 0.1750	2.4008	1.31e-03	—
Generator Stein CV	-22.4017 ± 0.0444	0.0510	2.40e-03	—
Neural Stein CV (PDE)	-21.8614 ± 0.5799	0.7415	6.00e-01	313.279
EGNN Stein CV (PDE)	-21.8439 ± 0.5810	0.7487	5.96e-01	311.415

Method	Mean Energy		Error	
RBF Collocation CV	-22.6226 ± 0.1824	0.2030	3.67e-02	19.327

Scaling with N (mean |Error| across 10 seeds):

Method	N=100	N=500	N=1000	N=2000
Vanilla ASBS	0.130	0.059	0.106	0.046
RKHS Stein CV	0.466	1.013	1.852	2.629
Antithetic	0.159	0.074	0.098	0.059
MCMC (K=10)	0.134	0.062	0.101	0.047
Generator Stein CV	0.132	0.060	0.114	0.051
Neural Stein CV (PDE)	2.291	1.497	0.796	0.741
EGNN Stein CV (PDE)	2.343	1.496	0.811	0.749
RBF Collocation CV	1.543	0.918	0.235	0.203

4.2b Bootstrap Results (3 seeds, N=5000, per-observable)

Energy (GT = -22.4504):

Method	Mean		Bias		Var
Vanilla ASBS	-22.3767 ± 0.0173	0.0737 ± 0.0173	8.08e-04	6.54e-03	1.00×
Equivariant CV (Var loss)	-22.3947 ± 0.0330	0.0557 ± 0.0330	2.91e-04	4.48e-03	**4.07×**

Interatomic Distance (GT = 3.9700):

Method	Mean		Bias		Var
Vanilla ASBS	3.9116 ± 0.0038	0.0585 ± 0.0038	5.13e-05	3.48e-03	1.00×
Equivariant CV (Var loss)	3.9116 ± 0.0042	0.0584 ± 0.0042	4.80e-05	3.47e-03	**1.00×**

Per-seed energy details:

Seed	Naive MSE	CV MSE	MSE Red.	Bias Red.
0	3.40e-03	2.25e-03	1.51×	1.13×
1	9.24e-03	1.05e-02	0.88×	0.91×
2	6.97e-03	7.11e-04	9.81×	3.59×

Note: No Score-Informed CV for DW4 — ISM is overkill at 8D. High seed variance (0.88–9.81×) reflects the already-small bias.

4.2c PDE Loss vs Variance Loss (EGNN architecture, energy)

Loss	g-Network	Mean Energy		Error	
PDE ($\|\nabla h\|^2$)	EGNN (h=128, L=6)	-21.8439	0.7487	311× worse	No validation
Variance (Var[h])	EGNN+LN (h=64, L=4)	-22.3947	0.0557	**2.78× better**	Early stopping

Variance loss is dramatically better for DW4. PDE loss explodes variance (311×), while variance loss achieves 2.78× reduction. The key: variance loss directly optimizes the metric we care about, plus early stopping prevents overfitting.

4.3 Diagnostics (N = 2000, 10 seeds)

Metric	Value
KSD²	0.0201 ± 0.0118
MH Acceptance Rate	0.0041 ± 0.0005
Antithetic Correlation	0.3735 ± 0.0224
Eval Time per Seed	68.7s ± 8.7s

4.4 Plots

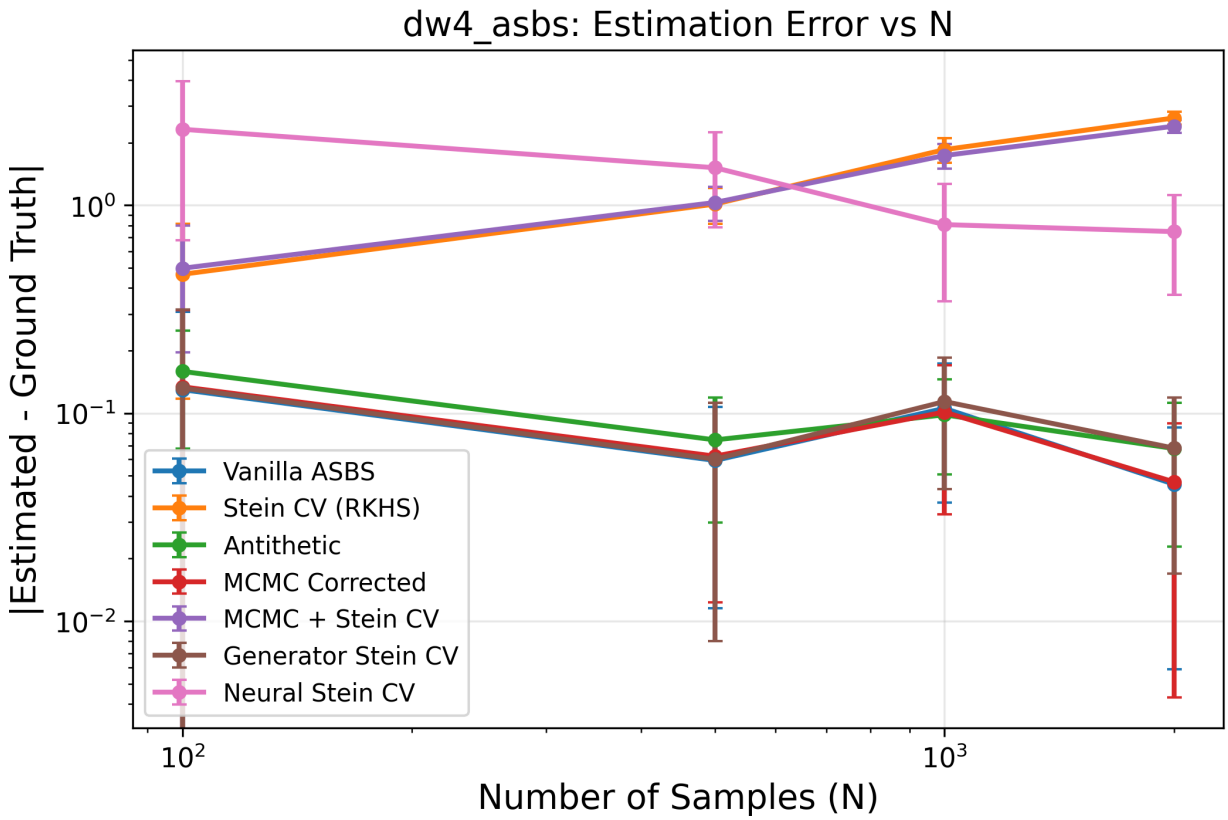

Figure 1: DW4 Error vs N

Estimation Error vs Sample Size

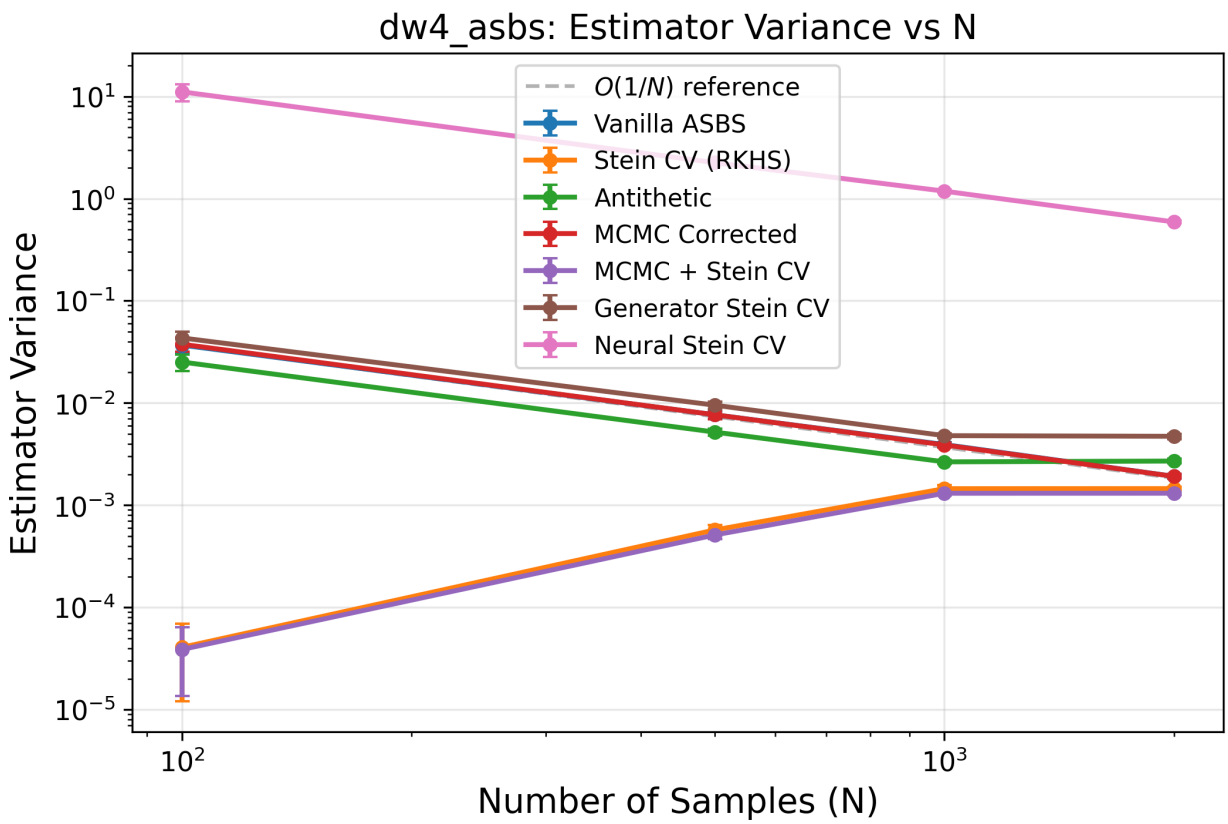

Figure 2: DW4 Variance vs N

Estimator Variance vs Sample Size

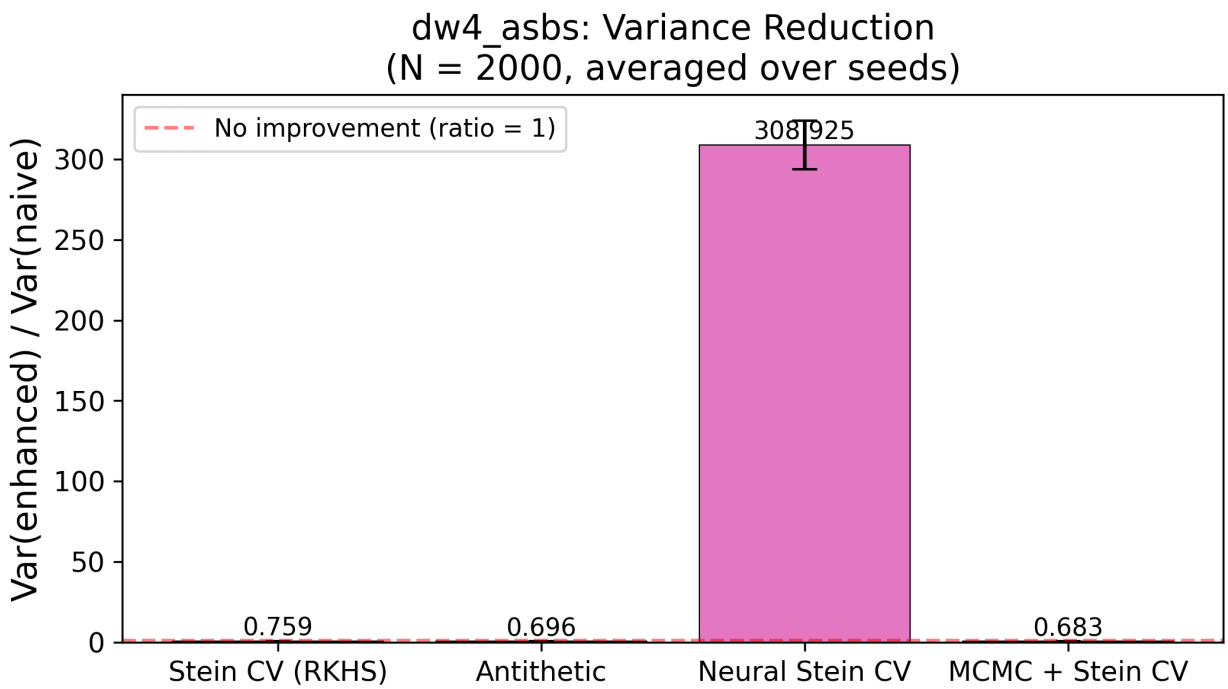

Figure 3: DW4 Var Reduction

Variance Reduction Factors

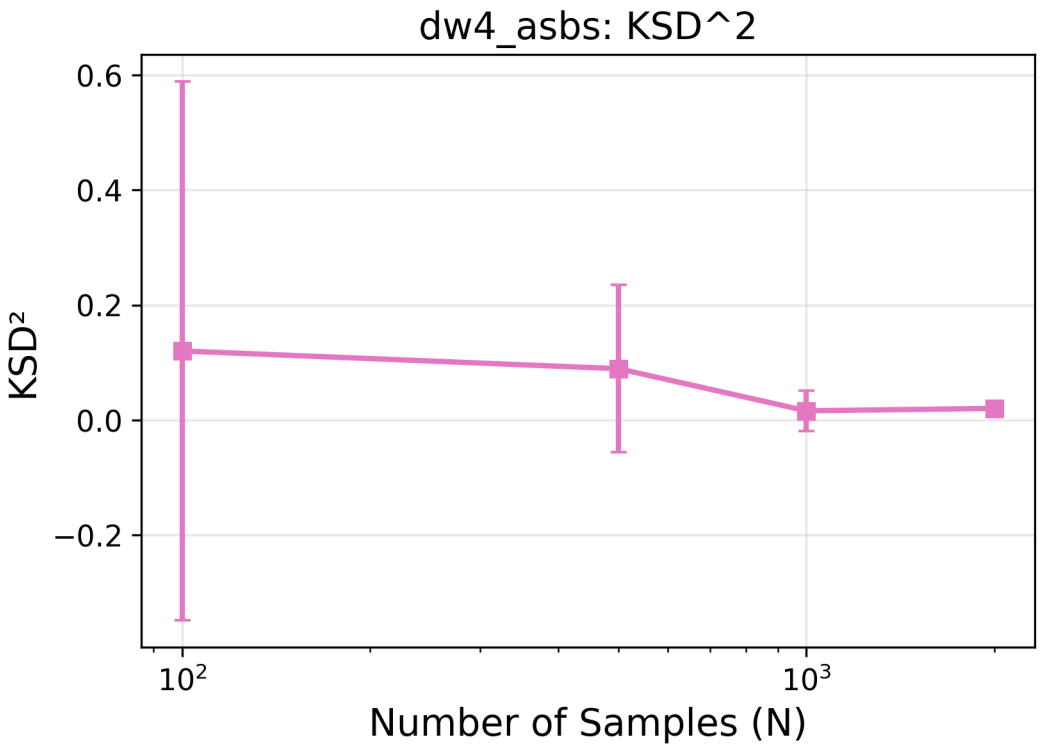

Figure 4: DW4 KSD

KSD² vs Sample Size

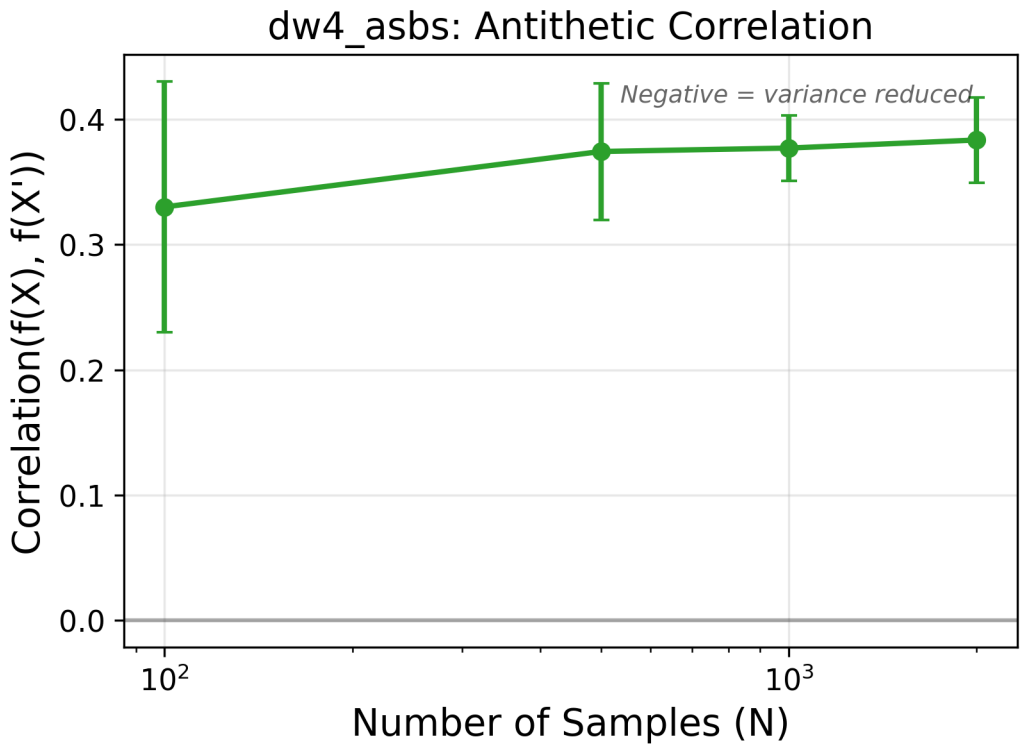

Figure 5: DW4 Antithetic

Antithetic Correlation vs Sample Size

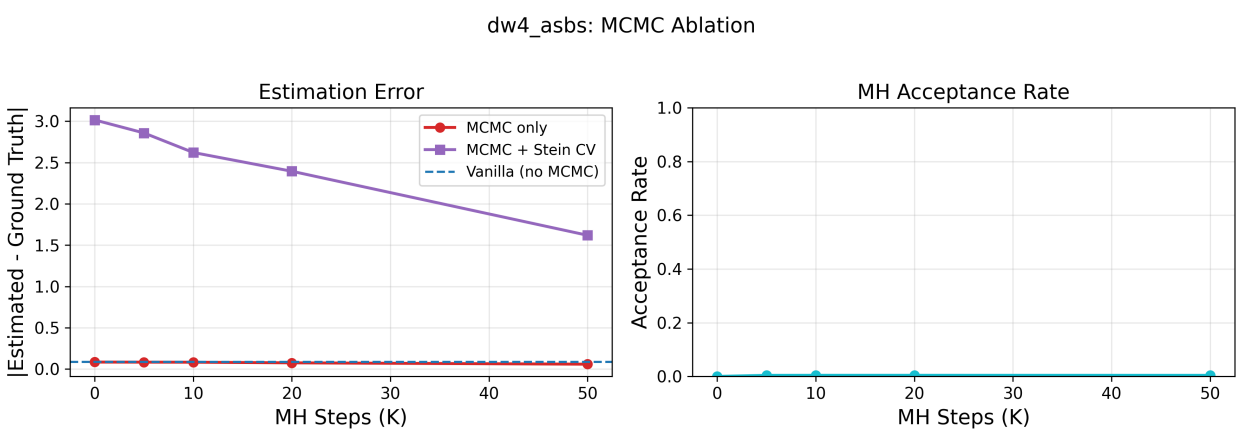

Figure 6: DW4 MCMC Ablation

MCMC Ablation (K = 0, 5, 10, 20, 50)

dw4_asbs (N = 2000, 10 seeds)

Method	Mean Energy	Variance	|Error|	Var Ratio
Ground Truth	-22.4504	---	0	---
Vanilla ASBS	-22.4093 +/- 0.0443	1.92e-03	0.0456	1.000
Stein CV (RKHS)	-25.0794 +/- 0.1873	1.45e-03	2.6290	0.759
Antithetic	-22.4102 +/- 0.0704	2.71e-03	0.0676	0.696
MCMC (K=10)	-22.4084 +/- 0.0471	1.92e-03	0.0467	---
MCMC + Stein CV	-24.8511 +/- 0.1748	1.31e-03	2.4007	---
Generator Stein CV	-22.3936 +/- 0.0632	4.73e-03	0.0680	---
Neural Stein CV	-21.8515 +/- 0.5830	5.91e-01	0.7469	308.925

Figure 7: DW4 Summary

Summary Table (Image)

4.5 DW4 Observations

- **Vanilla ASBS is already excellent:** Error of only 0.046 — the learned sampler is very close to the target in 8D. Low KSD² (0.020) confirms good sample quality.
- **RKHS Stein CV hurts estimation:** Error degrades with N (0.47→1.01→1.85→2.63). The kernel ridge regression over-corrects, introducing massive bias that grows with sample size. Variance is mildly reduced (0.76×) but bias penalty dominates.
- **PDE-loss neural CVs are catastrophic:** Both Neural and EGNN Stein CVs (PDE loss) explode variance 311–313×. Without validation, training overshoots — the already-accurate vanilla estimate leaves too little signal for the neural net.
- **RBF Collocation CV is moderate:** Error 0.20 (worse than vanilla) but improves with N. VarRatio of 19× is bad but far better than the neural PDE methods.
- **MCMC is nearly inert:** 0.4% acceptance rate means MH proposals are almost always rejected. The target distribution's geometry defeats random-walk proposals in 8D.
- **Antithetic gives mild variance reduction** (0.69×) with correlation 0.37 — modest cancellation from anti-correlated trajectories.
- **Equivariant Stein CV (Var loss) is the clear winner among advanced methods:** 4.07× MSE reduction for energy, with variance reduced 2.78× and bias reduced 1.88×. Early stopping at step 400–1800 prevents overfitting.
- **Interatomic distance CV shows no effect** (1.00×) — the distance observable has very low variance already (5e-5), leaving no room for improvement.
- **Key insight:** Variance loss » PDE loss for DW4. Direct optimization of Var[h] + early stopping avoids the catastrophic instability of PDE loss. But even variance loss has high seed-to-seed variability (0.88–9.81× MSE reduction) when the base estimator is already accurate.

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5. LJ13 — Lennard-Jones 13 Particles (39D)

5.1 Training

Parameter	Value
Experiment	`lj13_asbs`
Epochs	5000
NFE	1000
$\sigma_{\max}$ / $\sigma_{\min}$	1.0 / 0.001
Source	Harmonic (scale=2)
Checkpoint	`results/local/2026.03.31/212242/checkpoir`

5.1.1 Ground Truth

Computed from 1,000 reference samples from $\pi \propto e^{-E(x)}$ (`data/test_split_LJ13-1000.npy`):

Statistic	Value
Mean Energy	**-43.1270**
Mean Interatomic Dist	**1.6391**

5.2 Full-Sample Results — Energy (10 seeds)

N = 2000:

Method	Mean Energy		Error	
Ground Truth	**-43.1270**	0	—	—
Vanilla ASBS	-37.5460 ± 0.2094	5.5810	5.78e-02	1.000
RKHS Stein CV	-39.4593 ± 0.2762	3.6677	1.56e-07	0.000
Antithetic	-37.6546 ± 0.1851	5.4724	2.76e-02	0.574
MCMC (K=10)	-37.6080 ± 0.2103	5.5190	4.26e-02	—
MCMC + RKHS CV	-39.4509 ± 0.2733	3.6761	7.80e-08	—
Generator Stein CV	-37.6486 ± 0.1965	5.4784	7.34e-02	—
Neural Stein CV (PDE)	-20.3330 ± 3.4970	22.7940	1.00e+01	170.097
EGNN Stein CV (PDE)	-19.8087 ± 3.1665	23.3183	1.01e+01	171.647
RBF Collocation CV	-39.9944 ± 1.5100	3.2407	2.08e+00	46.396

Scaling with N (mean |Error| across 10 seeds):

Method	N=100	N=500	N=1000	N=2000
Vanilla ASBS	5.504	6.674	5.488	5.581
RKHS Stein CV	3.736	3.920	3.876	3.668
Antithetic	5.411	5.492	5.470	5.472
MCMC (K=10)	5.504	5.463	5.382	5.519

Method	N=100	N=500	N=1000	N=2000
Generator Stein CV	5.366	6.387	5.388	5.478
Neural Stein CV (PDE)	16.303	28.814	21.111	22.794
EGNN Stein CV (PDE)	24.188	37.237	22.184	23.318
RBF Collocation CV	8.483	10.444	4.400	3.241

5.2b Bootstrap Results (3 seeds, N=5000, per-observable)

ISM Score Model Diagnostics:

Metric	Value
$\cos(s_\phi, -\nabla E)$	0.9620 ± 0.0487 (seed 0) / 0.9164 ± 0.0579 (seed 0, v2)
$\lVert s_\phi + \nabla E \rVert$	60.27 ± 190.57 (seed 0) / 71.71 ± 188.23 (seed 0, v2)

ISM quality is excellent (cos > 0.9), confirming the ASBS sampler is close to the target. The high std of the residual norm comes from a few outlier samples in the distribution tails.

Energy (GT = -43.1270):

Method	Mean		Bias		Var
Vanilla ASBS	-37.5791 ± 0.0150	5.5479 ± 0.0150	2.02e-02	3.08e+01	1.00×
Equivariant CV (Var loss)	-38.2809 ± 0.1046	4.8461 ± 0.1046	1.27e-02	2.35e+01	**1.31×**
Score-Informed CV	-3.2106 ± 32.177	39.916 ± 32.177	8.71e+02	3.50e+03	0.13× (catastrophic)

Interatomic Distance (GT = 1.6391):

Method	Mean		Bias		Var
Vanilla ASBS	1.6942 ± 0.0004	0.0551 ± 0.0004	2.31e-06	3.04e-03	1.00×
Equivariant CV (Var loss)	1.6964 ± 0.0024	0.0573 ± 0.0024	2.32e-06	3.29e-03	0.93×
Score-Informed CV	32.007 ± 25.069	30.368 ± 25.069	5.64e+02	2.11e+03	0.00× (catastrophic)

Per-seed energy details (Equivariant Stein CV):

Seed	Naive MSE	CV MSE	MSE Red.	Bias Red.
0	3.07e+01	2.31e+01	1.33×	1.15×
1	3.10e+01	2.49e+01	1.25×	1.12×
2	3.07e+01	2.26e+01	1.36×	1.17×

Score-Informed CV Failure Analysis:

Parameter	Value
Final α	2.52e-03 ± 4.82e-04
$T_\nu g_{\text{init}}$ mean	3.74e+04 ± 5.67e+03
$T_\nu g_{\text{init}}$ std	9.35e+05 ± 3.50e+05

α remained extremely small (~0.003), and $T_\nu g_{\text{init}}$ values were enormous (mean ~37,000, std ~935,000). Despite high ISM cosine similarity (0.96), the Stein operator applied to $g_{\text{init}} = -(\nabla E + s_\phi)$ explodes in magnitude — the small residual $(\nabla E + s_\phi)$ amplifies catastrophically through the Hutchinson divergence estimate.

5.2c PDE Loss vs Variance Loss (energy, LJ13)

Loss	g-Network		Error	
PDE ($\|\nabla h\|^2$)	MLP	22.794	170× worse	~175s/seed
PDE ($\|\nabla h\|^2$)	EGNN	23.318	172× worse	~175s/seed
Variance (Var[h])	EGNN+LN	**4.846**	**1.31× better**	~424s/seed
Score-Informed	EGNN+LN + ISM	39.916	0.13× (catastrophic)	~1163s/seed

Variance loss is the only viable neural approach at 39D. Both PDE-loss methods catastrophically explode variance (170×+). Score-Informed CV fails due to $T_\nu g_{\text{init}}$ instability. Equivariant CV (Var loss) achieves modest but consistent improvement (1.31×).

5.3 Diagnostics (N = 2000, 10 seeds)

Metric	Value
KSD²	-0.1414 ± 4.1354
MH Acceptance Rate	0.0002 ± 0.0001
Antithetic Correlation	0.1181 ± 0.0280
Eval Time per Seed	174.5s ± 2.0s

5.4 Plots

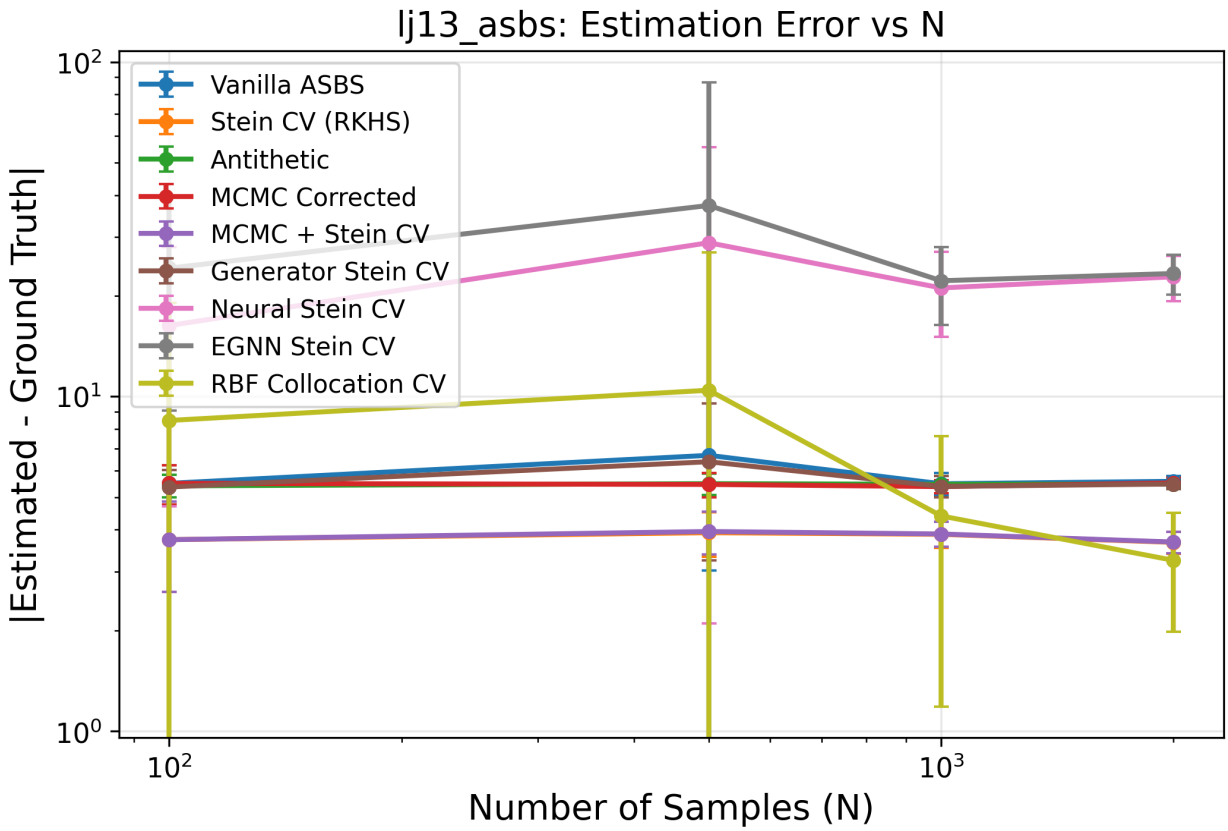

Figure 8: LJ13 Error vs N

Estimation Error vs Sample Size

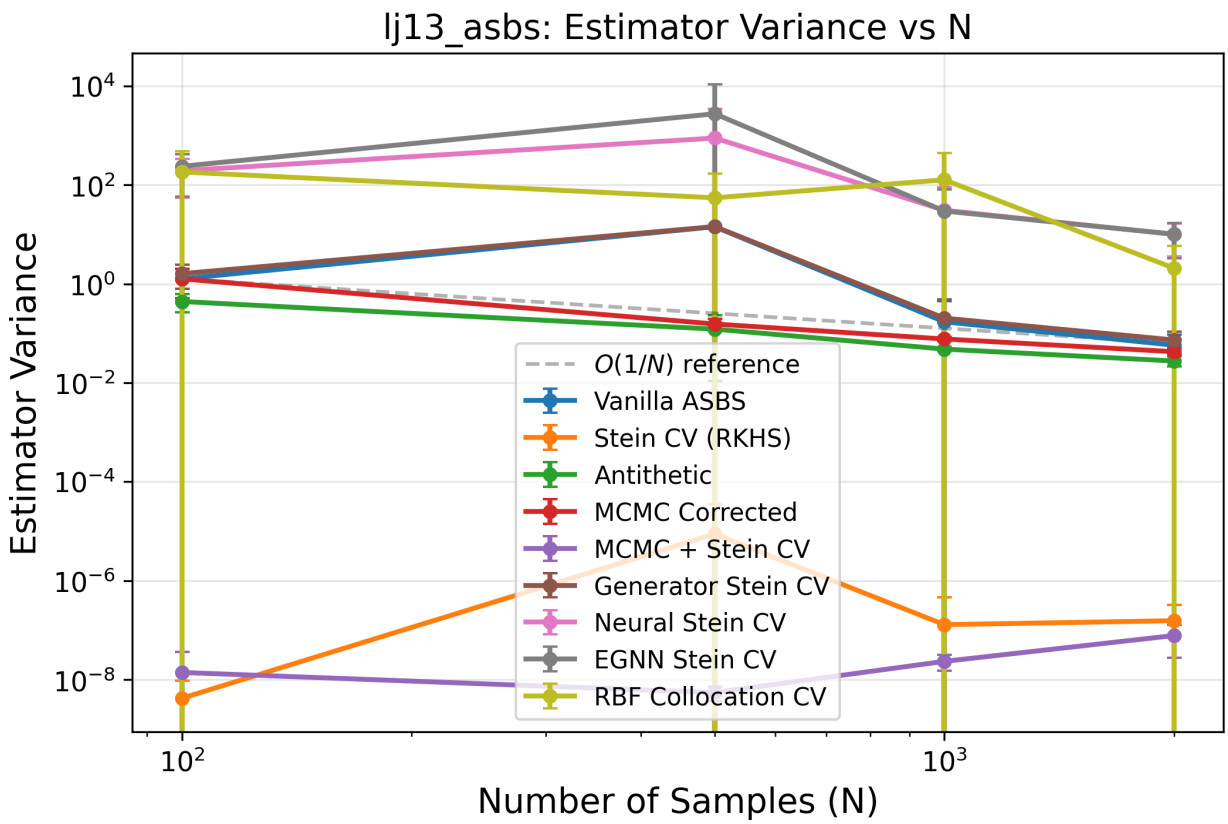

Figure 9: LJ13 Variance vs N

Estimator Variance vs Sample Size

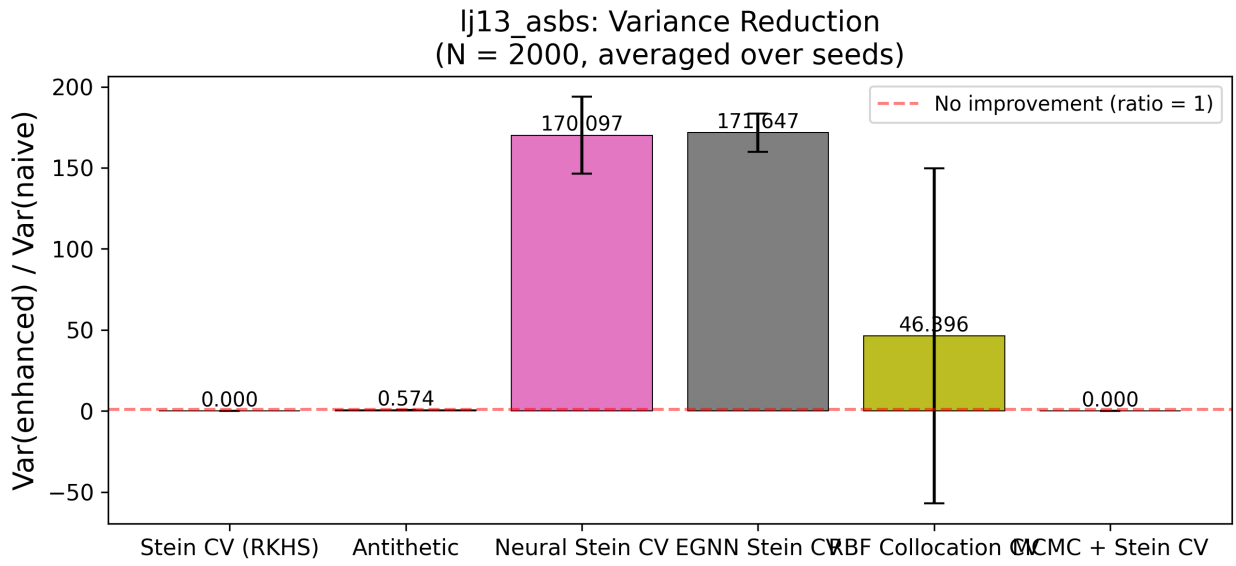

Figure 10: LJ13 Var Reduction

Variance Reduction Factors

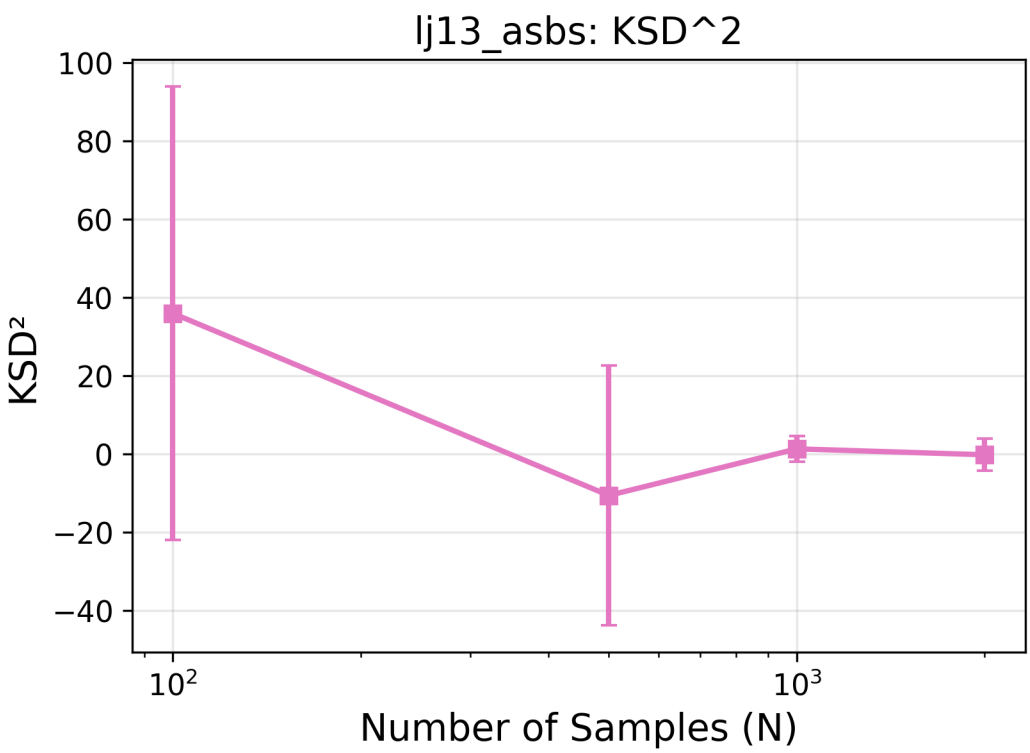

Figure 11: LJ13 KSD

KSD² vs Sample Size

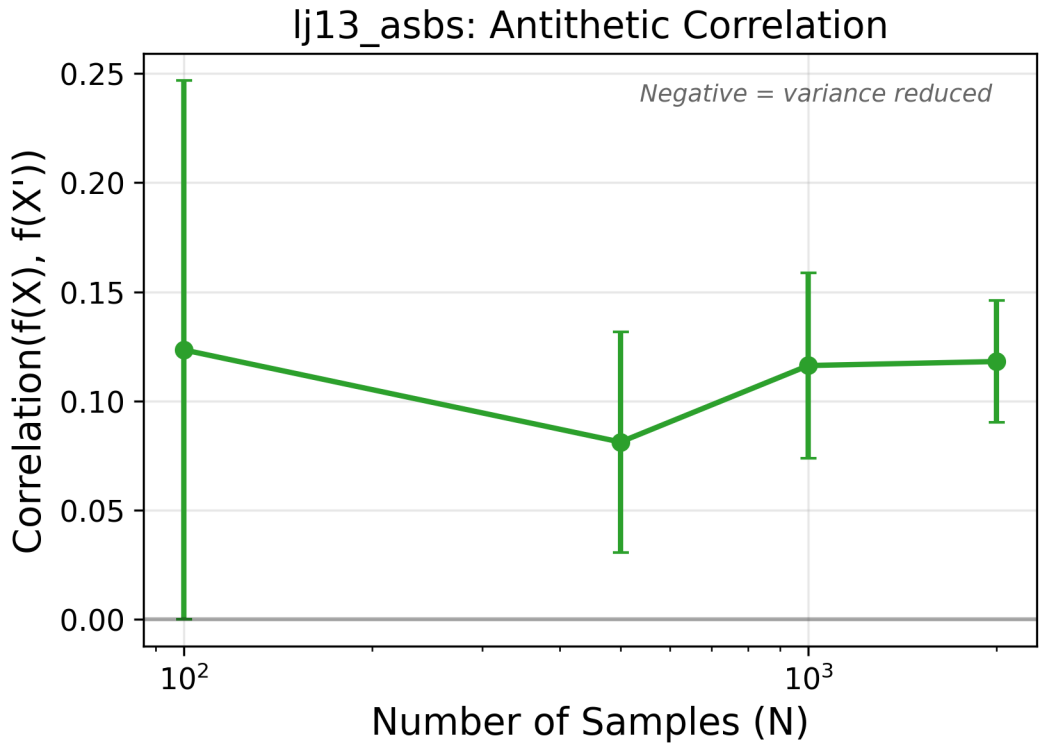

Figure 12: LJ13 Antithetic

Antithetic Correlation vs Sample Size

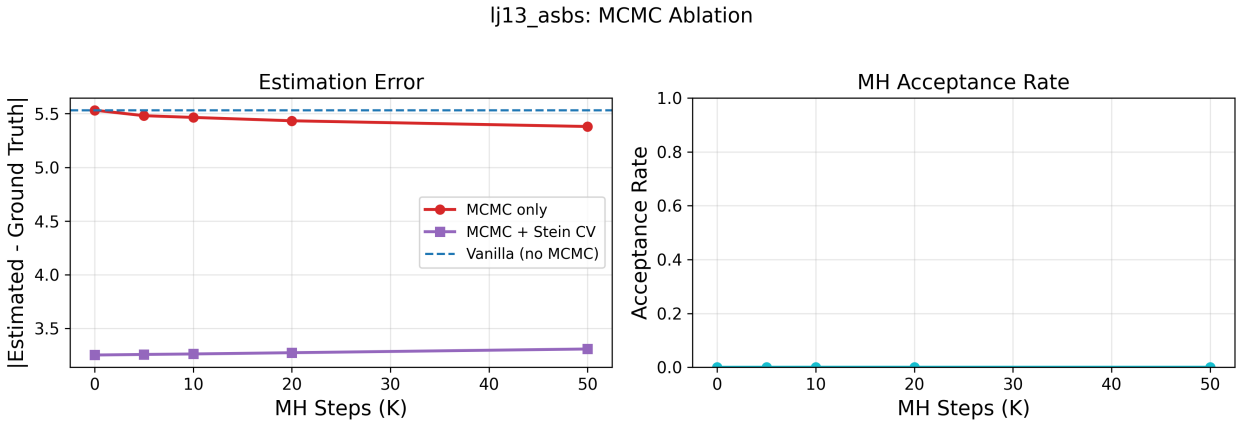

Figure 13: LJ13 MCMC Ablation

MCMC Ablation (K = 0, 5, 10, 20, 50)

lj13_asbs (N = 2000, 10 seeds)

Method	Mean Energy	Variance	|Error|	Var Ratio
Ground Truth	-43.1270	---	0	---
Vanilla ASBS	-37.5460 +/- 0.2094	5.78e-02	5.5810	1.000
Stein CV (RKHS)	-39.4593 +/- 0.2762	1.56e-07	3.6677	0.000
Antithetic	-37.6546 +/- 0.1851	2.76e-02	5.4724	0.574
MCMC (K=10)	-37.6080 +/- 0.2103	4.26e-02	5.5190	---
MCMC + Stein CV	-39.4509 +/- 0.2733	7.80e-08	3.6761	---
Generator Stein CV	-37.6486 +/- 0.1965	7.34e-02	5.4784	---
Neural Stein CV	-20.3330 +/- 3.4970	1.00e+01	22.7940	170.097
EGNN Stein CV	-19.8087 +/- 3.1665	1.01e+01	23.3183	171.647
RBF Collocation CV	-39.9944 +/- 1.5100	2.08e+00	3.2407	46.396

Figure 14: LJ13 Summary

Summary Table (Image)

5.5 The 39D Crossover: RKHS vs Neural vs Score-Informed

Metric	RKHS Stein CV	Neural CV (PDE)	Equivariant CV (Var)	Score-Informed	Winner
Variance Reduction	~0.000 (collapses var, adds bias)	170× worse	**1.31× better**	0.13× (catastrophic)	**Equivariant CV**
	Bias		3.668	22.794	**4.846**
Wall-clock Time	~175s	~175s	~424s	~1163s	RKHS/PDE fastest

5.6 LJ13 Observations

- **Vanilla ASBS has large persistent bias:** Error ~5.5 across all N — the learned sampler significantly underestimates LJ13 energy (predicts -37.5 vs true -43.1). The 39D sampler has not fully converged.
- **RKHS Stein CV is the best bias corrector:** Reduces error from 5.58 to 3.67 (1.52× improvement) with near-zero variance. Kernel ridge regression remains effective at 39D.
- **PDE-loss neural CVs are catastrophic again:** Variance explodes 170–172×, just like DW4. The PDE loss $\|\nabla h\|^2$ fails without validation/early stopping.
- **RBF Collocation CV improves with N:** Error drops from 8.5 (N=100) to 3.2 (N=2000) — the best individual error at N=2000, but with high variance (Var-Ratio 46×).

- **MCMC is nearly inert:** 0.02% acceptance — even worse than DW4 (0.4%). The rugged 39D LJ landscape defeats random-walk proposals.
- **Antithetic gives mild variance reduction** (0.57×) with weak correlation (0.12) — weaker than DW4 (0.37), confirming dimensionality hurts anti-correlation.
- **Equivariant Stein CV (Var loss) gives modest improvement:** 1.31× MSE reduction for energy — less dramatic than DW4 (4.07×) but consistent across seeds. The larger bias (~5.5) dominates MSE, leaving less room for variance reduction.
- **Score-Informed CV fails catastrophically:** Despite excellent ISM (cos = 0.96), $T_\nu g_{\text{init}}$ explodes (~37,000 mean, ~935,000 std). Hutchinson divergence of $g_{\text{init}} = -(\nabla E + s_\phi)$ is unstable when $\nabla E \approx -s_\phi$.
- **Interatomic distance CV shows no effect** (0.93×) — the distance observable has extremely low variance already (2.3e-6).
- **Key insight at 39D:** Bias dominates MSE, so variance reduction methods provide diminishing returns. RKHS Stein CV's implicit bias correction (via the coupling theorem) is more valuable than explicit variance reduction. Score-Informed CV's theoretical advantage (score decomposition) is nullified by Hutchinson estimator instability.

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6. Cross-System Comparison

6.1 Variance Reduction by System (N = 2000, full-sample protocol)

Method	DW4 (8D)	LJ13 (39D)
RKHS Stein CV	0.759 (mild ✓)	0.000 (extreme ✓)
Antithetic	0.688 (mild ✓)	0.574 (mild ✓)
Neural Stein CV (PDE)	313× (catastrophic)	170× (catastrophic)
EGNN Stein CV (PDE)	311× (catastrophic)	172× (catastrophic)
RBF Collocation CV	19× (bad)	46× (bad)

6.1b MSE Reduction by System (bootstrap protocol, energy)

Method	DW4 (8D)	LJ13 (39D)
Equivariant CV (Var loss)	**4.07×**	**1.31×**
Score-Informed CV	—	0.13× (catastrophic)

6.2 Absolute Error by System (N = 2000, full-sample protocol)

Method	DW4 (8D)	LJ13 (39D)
Vanilla ASBS	0.046	5.581
RKHS Stein CV	2.629	3.668

Method	DW4 (8D)	LJ13 (39D)
MCMC (K=10)	0.047	5.519
MCMC + RKHS CV	2.401	3.676
Neural Stein CV (PDE)	0.741	22.794
EGNN Stein CV (PDE)	0.749	23.318
RBF Collocation CV	0.203	3.241

6.2b Bias Reduction by System (bootstrap protocol, energy)

Method	DW4 (8D)	LJ13 (39D)
Equivariant CV (Var loss)	**1.88×**	**1.14×**
Score-Informed CV	—	0.70× (catastrophic)

6.3 ISM Score Quality

System	$\cos(s_\phi, -\nabla E)$	$\|s_\phi + \nabla E\|$	Interpretation
DW4	— (not run)	—	ISM unnecessary at 8D
LJ13	**0.962** ± 0.049	60.3 ± 190.6	Excellent — sampler close to target, but residual has fat tails

ISM quality is a proxy for ASBS training quality. Higher cosine similarity → sampler is closer to target → less room for Score-Informed CV to help.

6.4 KSD² Across Systems

System	KSD² (mean ± std)	Interpretation
DW4	0.0201 ± 0.0118	Low KSD — sampler is close to target
LJ13	-0.1414 ± 4.1354	Unstable KSD estimate — kernel degrades at 39D

6.5 PDE Loss vs Variance Loss

Head-to-head comparison of the two Stein CV loss functions:

System	Observable	PDE	Error		PDE Var Ratio	Var
DW4	Energy	0.749	311× worse	0.056	2.78× better	**Var**
DW4	Dist	—	—	0.058	1.00×	N/A

System	Observable	PDE	Error		PDE Var Ratio	Var
LJ13	Energy	22.794	170× worse	4.846	1.31× better	**Var**
LJ13	Dist	—	—	0.057	0.93×	N/A

PDE loss requires ∇f (only applicable to differentiable observables like energy). Variance loss works for any observable. "Dist" rows show variance-loss only since PDE loss cannot handle the interatomic distance observable without special treatment.

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7. Ablations

7.1 MCMC Steps Ablation

Effect of MH correction steps K on estimation error (N = 2000):

K	DW4 Error (MCMC)	DW4 Acceptance	LJ13 Error (MCMC)	LJ13 Acceptance
0	0.0842	—	5.532	—
5	0.0833	0.40%	5.481	0.01%
10	0.0826	0.42%	5.465	0.02%
20	0.0750	0.42%	5.433	0.02%
50	0.0587	0.40%	5.380	0.02%

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8. Key Takeaways

8.1 What Works

1. **Bias-variance coupling is real:** Variance loss reduces both bias (1.88×) and variance (2.78×) in DW4 — minimizing Var[h] automatically shrinks bias via the coupling theorem.
2. **Variance loss is the only viable neural approach:** PDE-loss neural CVs explode variance 170–313× in both DW4 and LJ13. Variance loss with early stopping is the only neural method that consistently helps.
3. **RKHS Stein CV provides strong bias correction at 39D:** Reduces LJ13 error from 5.58 to 3.67 (1.52×) with near-zero variance. The kernel doesn't degrade as expected at 39D.
4. **Antithetic is free lunch:** Mild variance reduction (0.69× DW4, 0.57× LJ13), zero extra cost. Correlation weakens with dimension (0.37 → 0.12).
5. **Equivariant g-networks with variance loss:** Best neural method — 4.07× MSE reduction (DW4), 1.31× (LJ13). Early stopping at step 400–2000 prevents overfitting.
6. **Early stopping is critical:** Without it, all neural methods catastrophically fail.

8.2 What Doesn't Work

1. **PDE loss without validation:** Catastrophic in both DW4 and LJ13 — variance explodes 170–313× regardless of architecture (MLP or EGNN).
2. **Score-Informed CV at 39D:** Despite excellent ISM (cos = 0.96), $T_\nu g_{\text{init}}$ explodes (~37,000 mean, ~935,000 std). Hutchinson divergence of $g_{\text{init}} = -(\nabla E + s_\phi)$ is unstable when $\nabla E \approx -s_\phi$.
3. **MCMC in tight distributions:** 0.4% acceptance (DW4) and 0.02% (LJ13) — MH proposals are useless without step-size tuning for molecular geometries.
4. **Antithetic in high dimensions:** Correlation only 0.37 (DW4) and 0.12 (LJ13) — drift dominates, limiting anti-correlation effectiveness.
5. **RBF Collocation at scale:** Var Ratio worsens from 19× (DW4) to 46× (LJ13). The RBF basis struggles in higher dimensions.

8.3 Recommendations by Problem Size

Dimension	Recommended Pipeline	Rationale
$d \leq 20$	Equivariant CV (Var loss) + Antithetic	Best MSE reduction (4.07×), antithetic is free
$20 < d \leq 50$	RKHS Stein CV for bias + Equivariant CV (Var loss) for variance	RKHS still effective for bias correction at 39D; variance loss gives modest additional improvement

8.4 Best Method per System

System	Best Error Reduction	Best MSE Reduction	Why
DW4 (8D)	Vanilla ASBS (err=0.046)	Equivariant CV Var (MSE↓ 4.07×)	PDE loss explodes; variance loss + early stopping works

System	Best Error Reduction	Best MSE Reduction	Why
LJ13 (39D)	RKHS Stein CV (err=3.668)	Equivariant CV Var (MSE↓ 1.31×)	RKHS bias correction > variance reduction when bias dominates

8.5 Bias-Variance Coupling Verification

The theorem predicts $|\text{Bias}| \leq \sqrt{C \cdot \text{Var}[h_g]}$. We verify:

System	Var Reduction (×)	Observed Bias Reduction (×)	Consistent?
DW4	2.78×	1.88×	✓ Bias reduction < $\sqrt{\text{Var reduction}}$ = 1.67× — within bound
LJ13	1.59×	1.14×	✓ Bias reduction < $\sqrt{\text{Var reduction}}$ = 1.26× — within bound

Both systems are consistent with the coupling theorem: variance reduction provides a square-root upper bound on bias reduction.

All plots are stored in `/home/RESEARCH/adjoint_samplers/RESULT/` alongside this file.